

ss Cheat Sheet

See which ports are open, which program owns them, and who is connected. The modern replacement for netstat.

Using it

```
ss [options] [state STATE] [filter] ← no options = open connections (not listening ports)
sudo ss -tulpn ← the classic: every listening TCP/UDP port, with numbers and programs
```

Option	Means	Option	Means
-t / -u	TCP / UDP sockets	-p	Program using it (sudo to see all)
-l	Listening sockets only (servers)	-4 / -6	IPv4 only / IPv6 only
-a	All: listening and connected	-s	Summary counts per protocol
-n	Numbers: 22 not ssh	-H	No header line (good for scripts)

Reading the output of sudo ss -tulpn

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.1:323	0.0.0.0:*	users:(("chronyd",pid=644,fd=5))
tcp	LISTEN	0	128	0.0.0.0:22	0.0.0.0:*	users:(("ssh",pid=812,fd=3))
tcp	LISTEN	0	244	127.0.0.1:5432	0.0.0.0:*	users:(("postgres",pid=990,fd=6))
tcp	LISTEN	0	128	:::22	:::*	users:(("ssh",pid=812,fd=4))
tcp	LISTEN	0	511	:::80	:::*	users:(("nginx",pid=1020,fd=6))

Column	Means
Netid / State	Protocol, then state: LISTEN = waiting for connections; UDP always shows UNCONN (normal)
Recv-Q / Send-Q	Data waiting in queues (on LISTEN: queued connections / max). Near 0 = fine; growing = stuck
Local Address:Port	The address and port this machine listens on (see below)
Peer Address:Port	The other end. * = anyone (listeners have no peer yet)
Process	Program name, process ID and file descriptor. Empty without sudo

Local address	Reachable from	Local address	Reachable from
0.0.0.0:22	Every IPv4 address on the host	127.0.0.1:5432	This machine only (safe)
:::22 / *:80	Every address (IPv6 and usually IPv4)	192.168.1.50:80	Only that one address

Connection states

State	Means	State	Means
LISTEN	A server waiting for connections	TIME-WAIT	Just closed; clears itself in ~60 s
ESTAB	Connected and talking	CLOSE-WAIT	Peer hung up, program hasn't (app bug)
SYN-SENT	Trying to connect; stuck = firewall	UNCONN	A UDP socket (normal)

Filtering

Command	Shows
ss -tn state established	Only established TCP connections (the State column is then left out)
sudo ss -tlnp sport = :22	Who is listening on port 22 (sport = local port)
ss -tn '(dport = :443 or dport = :80)'	Connections out to web servers (dport = remote port)
ss -tn dst 203.0.113.5	Connections to one remote host (src = local address)

Handy recipes

Command	Does
ss -s	Quick totals: TCP connections, how many in TIME-WAIT
ss -Htn state established wc -l	Counts current TCP connections
sudo ss -tnp sport = :22	Who is connected over SSH right now

Remember: `sudo ss -tulpn` answers "what is listening, and who owns it?" Old habits translate directly: `netstat -tulpn` becomes `ss -tulpn`, and `netstat -r` becomes `ip route`.